Linear Diophantine Representation Systems

Extension I

Scaling to Small Numbers · Scaled Digital Root · 3D Extension $pA + qB + rC$

Bilal el Issaoui · Amsterdam · 2026

> *"The digital root is scale-invariant.*
> *Every number — however small — is reachable via its scaled counterpart."*

This document is a formal extension of the main work *Linear Diophantine Representation Systems: $N = 19A + 9B$*. It covers three new subjects: (1) a formal definition of the scaled digital root that makes the system valid for all positive rational numbers, (2) worked examples for small numbers such as 42, 24, 18, 6, 3 and 1, and (3) a general theory for the three-dimensional extension $N = pA + qB + rC$.

§1 Scaling and Small Numbers

1.1 The problem: the Frobenius boundary

The main theorem A_0 = dr(N) holds for all N >= pq. For the pair (19, 9) we have pq = 171. Small numbers such as 42, 24 or 18 fall outside this range: the representation 19 * dr(N) + 9 * B would yield a negative B. This appeared to be a hard boundary of the system. The scaling extension introduced here shows that this boundary is not a limitation but a scaling effect.

1.2 Scale-invariance of the digital root

The digital root is invariant under multiplication by 10:

$$dr(N) = dr(10N) = dr(100N) = dr(10^k * N) \text{ for all } k >= 0$$

Proof: multiplying by 10 appends a zero to N. The digit sum is unchanged: S(10N) = S(N). Hence dr(10N) = dr(N). By induction this holds for every power of 10. QED

1.3 The scaled A_0 value

Let N be a positive rational number with N < pq. Choose the smallest k in N such that N' = 10^k * N >= pq. Compute the representation of N' in the standard system. The scaled representation of N is then:

$$A_0(N) = dr(N) / 10^k \quad B_0(N) = (N' - 19 * dr(N')) / (9 * 10^k)$$

The anchor value remains dr(N) — scaled by the same factor 10^k. The echo of the digital root is therefore structurally present at every scale.

1.4 Definition: scaled digital root

Definition 1.1 — Scaled digital root

Let N be a positive rational number and k the smallest non-negative integer such that N' = 10^k * N is an integer with N' >= pq. Then the **scaled digital root** of N is:

dr(N) = dr(N') / 10^k*

For integers N >= pq we have dr*(N) = dr(N). The scaled digital root extends the system to all positive rationals.

1.5 Example: 420 to 42 to 4.2

dr(420) = dr(42) = dr(4.2) = 6. The scaling chain shows how the representation scales proportionally:

N	k	N' = 10k * N	A0 = dr(N')/10k	B0	Verification
4200	0	4200	6	434	19*6 + 9*434 = 4200 ok
420	0	420	6	34	19*6 + 9*34 = 420 ok
42	1	420	0.6	3.4	19*0.6 + 9*3.4 = 42 ok
4.2	2	420	0.06	0.34	19*0.06 + 9*0.34 = 4.2 ok

The digital root echoes visibly at every scale: the anchor value 6 is always present, multiplied by the appropriate scaling factor.

1.6 Example: 2520 to 252 to 25.2

N = 2520 has dr = 9 and symmetry point (90, 90). The scaling chain shows how the symmetry point moves with the scale:

N	k	A0	B0	Symmetry point	Verification
2520	0	9	261	(90, 90)	19*90 + 9*90 = 2520 ok
252	1	0.9	26.1	(9, 9)	19*9 + 9*9 = 252 ok
25.2	2	0.09	2.61	(0.9, 0.9)	19*0.9 + 9*0.9 = 25.2 ok

The symmetry point (A = B) is preserved under scaling: if A = B for N', then A/10 = B/10 for N'/10.

§2 Small Numbers: 42, 24, 18, 6, 3 and 1

For each small number N: find the digital root, use it as A_0 at the appropriate scale, and compute B_0 via the standard formula. All computations are carried out via $N' = 10^k * N$.

N	dr(N)	k	N'	A0 = dr/10k	B0	Echo dr visible?
42	6	1	420	0.6	3.4	6 -> 0.6 ok
24	6	1	240	0.6	1.4	6 -> 0.6 ok
18	9	1	180	0.9	0.9	9 -> 0.9 ok
6	6	2	600	0.06	0.54	6 -> 0.06 ok
3	3	2	300	0.03	0.27	3 -> 0.03 ok
1	1	2	190	0.01	0.19	1 -> 0.01 ok

Verification table via N':

N	N'	dr(N')	A0(N')	B0(N')	19*A0 + 9*B0 = N'
42	420	6	6	34	19*6 + 9*34 = 114 + 306 = 420 ok
24	240	6	6	14	19*6 + 9*14 = 114 + 126 = 240 ok
18	180	9	9	9	19*9 + 9*9 = 171 + 9 = 180 ok
6	600	6	6	54	19*6 + 9*54 = 114 + 486 = 600 ok
3	300	3	3	27	19*3 + 9*27 = 57 + 243 = 300 ok
1	190	1	1	19	19*1 + 9*19 = 19 + 171 = 190 ok

2.1 The computation of 24 via 240 - 114

The step 240 - 114 follows directly from the standard formula:

$$A_0 = dr(240) = 6$$
$$19 \times 6 = 114$$
$$240 - 114 = 126$$
$$B_0 = 126 / 9 = 14$$
$$\text{Verification: } 19*6 + 9*14 = 114 + 126 = 240 \text{ ok}$$
$$\text{Scaled: } A_0(24) = 0.6 \; B_0(24) = 1.4$$

*The step 240 - 114 is simply N' - p * dr(N'), the numerator of the B_0 formula. This is the standard procedure, here made visible for a small number via scaling.*

2.2 The echo: why the digital root is always visible

The echo of the digital root across scales is not a coincidence. It is a direct consequence of scale-invariance (Theorem 1.2): because dr(N) = dr(10N) for all N, the anchor value A_0 = dr(N') always equals dr(N) regardless of the scale factor chosen. Dividing by 10^k does not change the proportional structure of the representation — it only rescales all coordinates uniformly.

$$dr^*(N) = dr(N) / 10^k$$

A_0 echoes dr(N) at every scale.

§3 Three-Dimensional Extension: $N = pA + qB + rC$

3.1 Motivation

The two-dimensional system $N = pA + qB$ with $p = 1 \pmod q$ has one central structural property: $A_0 = dr_q(N)$. The question is whether a third variable C can be added such that this property is preserved, and what conditions the triple (p, q, r) must satisfy.

3.2 Conditions for the triple (p, q, r)

Definition 3.1 — Valid 3D system

A triple (p, q, r) of positive integers is called a **valid 3D representation system** if the following conditions hold:

(V1) $p = 1 \pmod q$ — the modular base property
(V2) $\gcd(p, q) = 1$ — coprimality of p and q
(V3) r divides q — $r \mid q$, so r and q are modularly compatible
(V4) $p = 1 \pmod r$ — the modular property also holds for r

Under these conditions $A_0 = dr_q(N)$ still holds as the minimal A-coefficient.

3.3 Why r | q is necessary

In the 2D system: $N = pA \pmod q$ because $p = 1 \pmod q$. This gives A_0 directly as N mod q. When rC is added, the congruence structure must be preserved. If $r \mid q$ then $rC = 0 \pmod q$ for every C. Hence $N = pA \pmod q$ still holds — the C variable does not disturb the anchor value.

3.4 Structure theorem for 3D

Theorem 3.2 — Minimal A-coefficient in 3D

Let (p, q, r) be a valid 3D system (V1-V4) and N >= pq. Then for the system $N = pA + qB + rC$ with A, B, C >= 0:

$$A_0 = dr_q(N)$$

The minimal A-value is independent of the choice of C. The B and C coefficients distribute the remainder ($N - p * A_0$) between them, but do not change the anchor value A_0.

Proof (sketch):

From N = pA + qB + rC mod q: N = pA + rC (mod q). Because r | q we have rC = 0 (mod q) for every C. Hence N = pA = A (mod q), exactly as in the 2D case. The minimal A is therefore A_0 = N mod q = $dr_q(N)$. QED

3.5 The canonical example: (19, 9, 3)

The most natural 3D system for (19, 9) is (19, 9, 3), because 3 divides 9 and 19 = 1 (mod 3). The third variable C subdivides the 9B part further into steps of 3.

$$N = 19A + 9B + 3C \text{ with } A_0 = dr(N)$$

N	dr(N)	A0	Example (A, B, C)	Verification
171	9	9	(9, 0, 0)	19*9 = 171 ok
240	6	6	(6, 14, 0)	114 + 126 = 240 ok
240	6	6	(6, 11, 9)	114 + 99 + 27 = 240 ok
240	6	6	(6, 8, 18)	114 + 72 + 54 = 240 ok
420	6	6	(6, 34, 0)	114 + 306 = 420 ok
420	6	6	(6, 25, 27)	114 + 225 + 81 = 420 ok
958	4	4	(4, 98, 0)	76 + 882 = 958 ok
958	4	4	(4, 95, 9)	76 + 855 + 27 = 958 ok

The A0 value equals dr(N) in every case. The C variable redistributes the remainder in steps of 3, while B moves in steps of 9. Every row with the same A0 is a valid representation.

3.6 Representation count in 3D

In the 2D system R(N) = floor((floor(N/p) - A0) / q) + 1. In the 3D system the number of representations increases: for every valid pair (B, C) with qB + rC = N - p * A0 - ... there exists a representation. The asymptotic growth is:

$$R_3(p, q, r; N) \sim N^2 / (2 * p * q * r)$$

A precise closed formula for R3 is an open problem requiring further investigation.

3.7 Classification of 3D systems

Class	Conditions	Example	A0 = dr(N)?
3D-I	p=1(mod q), r|q, p=1(mod r)	(19, 9, 3)	Yes

3D-II	3D-I + phi(p) = 2q	(19, 9, 3)	Yes
3D-III	3D-II + p+q is a perfect number	(19, 9, 3)	Yes
3D-IV	3D-III + closed triangle + Matryoshka	(19,9,3) only?	Conjecture

Whether (19, 9, 3) is the only Class-3D-IV system is an open problem — analogous to the uniqueness of (19, 9) in the 2D case.

§4 Summary of New Theorems

Label	Statement	Status
U1.1	dr(N) = dr(10k * N) for all k >= 0	Proved
U1.2	dr*(N) = dr(N') / 10k (scaled digital root)	Definition
U1.3	A0(N) = dr*(N) for all N in Q+ via scaling	Theorem
U2.1	Symmetry point scales: (A, A) -> (A/10, A/10)	Corollary
U3.1	In 3D: A0 = drq(N) if r|q and p=1(mod r)	Theorem 3.2
U3.2	R3(N) ~ N^2 / (2pqr) asymptotically	Observation
V3	Is (19, 9, 3) the only 3D Class-IV system?	Open problem